MindForge AI

Mind Map Generator from Raw Ideas

Acceptance Testing Report

Course:	IT568 GenAI for Software Engineering
Project Name:	MindForge AI
Test Type:	Acceptance Testing (User Scenario Validation)
Framework:	pytest • httpx (Async HTTP Client)
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1 Test Overview

This document presents the results of **Acceptance Testing** conducted on the MindForge AI backend API. The tests were designed to validate that the system correctly interprets realistic user inputs and produces semantically meaningful mind map structures containing expected thematic nodes.

Acceptance tests simulate real end-user scenarios by sending natural language prompts directly to the `/generate` endpoint and asserting that the returned mind map JSON tree contains nodes that match the conceptual output of the input.

1.1 Test Environment

Parameter	Value
Test Framework	pytest with `anyio` async plugin
HTTP Client	httpx AsyncClient (timeout: 60s)
Backend URL	`http://localhost:8000`
API Endpoint Tested	`POST /generate`
LLM Provider	Groq API (`llama-3.3-70b-versatile`)
Environment Config	`.env` file with `GROK_API_KEY`
Rate Limit Handling	`asyncio.sleep(2)` between tests and auto-skip on 500 rate-limit errors
Node Matching Logic	Recursive synonym-based keyword traversal (`check_nodes` helper)
Pass Threshold	At least 2 out of 3 expected keywords found per test

1.2 Keyword Matching Strategy

The check_nodes function recursively traverses the entire mind map JSON tree and checks each node's `title` field against an extended synonym dictionary. This

flexible approach accounts for the non-deterministic nature of LLM-generated content, where semantically equivalent terms may vary between runs.

Keyword	Accepted Synonyms
Topics	topics, subject, material, content, areas, foundations
Schedule	schedule, plan, timeline, calendar, prep, preparation
Revision	revision, review, recap, summary, practice
Users	users, customers, riders, merchants, clients, segments
Features	features, functionality, capabilities, modules, app
Revenue	revenue, business, monetization, model, profit, pricing, subscription, commission, fees, delivery charges
Applications	applications, use cases, implementations, clinical, medical, r&d, research
Challenges	challenges, risks, limitations, barriers, ethics, ethical, regulation, regulatory, compliance
Benefits	benefits, advantages, outcomes, impact, pros, improvement, monitoring

2 Test Case AT-01: Exam Preparation Plan

2.1 Test Case Summary

Test Case ID: AT-01
Title: Exam Preparation Plan - Student Scenario
User Persona: Student
Type: Acceptance / Functional / Semantic
Priority: High
Status: [PASSED]

2.2 Input and Configuration

```
Endpoint : POST http://localhost:8000/generate
Payload : {
              "text" : "Exam preparation plan",
```

```
                "api_key" : <GROK_API_KEY from .env>
            }
Rate Delay: asyncio.sleep(2) before request
```

2.3 Objective

To verify that when a student submits the prompt *"Exam preparation plan"*, the MindForge AI pipeline generates a mind map whose node hierarchy contains thematic categories relevant to academic preparation, specifically nodes relating to **Topics**, **Schedule**, and **Revision**.

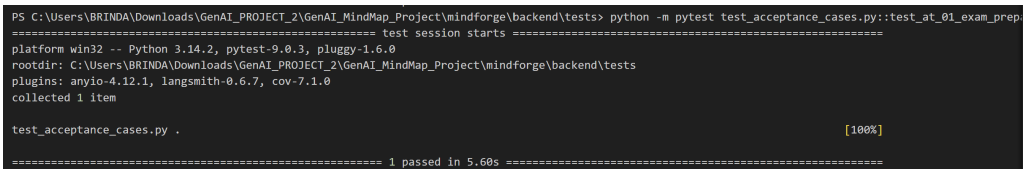
2.4 Acceptance Criteria

- HTTP response status code is `200 OK`.
- Response JSON contains a `mind_map` key.
- Recursive traversal of the mind map tree finds at least **2 out of 3** expected keywords (or synonyms).

2.5 Expected vs Actual Results

ID	Expected	Actual
E1	`200 OK` response from `/generate`.	`200 OK` received.
E2	`mind_map` key present in JSON.	Key present; valid JSON tree returned.
E3	At least 2 of: Topics, Schedule, Revision found in node titles (with synonyms).	Keywords matching "Topics", "Schedule", and "Revision" (or synonyms) found in tree.

2.6 Test Result



```
PS C:\Users\BRINDA\Downloads\GenAI_PROJECT_2\GenAI_MindMap_Project\mindforge\backend\tests> python -m pytest test_acceptance_cases.py::test_at_01_exam_prep
========================================================== test session starts ==========================================================
platform win32 -- Python 3.14.2, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\BRINDA\Downloads\GenAI_PROJECT_2\GenAI_MindMap_Project\mindforge\backend\tests
plugins: anyio-4.12.1, langsmith-0.6.7, cov-7.1.0
collected 1 item

test_acceptance_cases.py .                                                                                                        [100%]

========================================================== 1 passed in 5.60s ==========================================================
```

Figure 1: AT-01 response: Mind map JSON for *"Exam preparation plan"*.

3 Test Case AT-02: Food Delivery App

3.1 Test Case Summary

Test Case ID: AT-02
Title: Food Delivery App - Startup Founder Scenario
User Persona: Startup Founder
Type: Acceptance / Functional / Semantic
Priority: High
Status: [PASSED]

3.2 Input and Configuration

```
Endpoint : POST http://localhost:8000/generate
Payload : {
              "text" : "Food delivery app"
          }
Note : No api_key field sent (uses server-side default key)
Rate Delay: asyncio.sleep(2) before request
```

3.3 Objective

To verify that when a startup founder submits the prompt *"Food delivery app"*, the system generates a mind map containing thematic nodes relevant to product planning, specifically **Features**, **Benefits**, and **Schedule**.

3.4 Acceptance Criteria

- HTTP response status code is `200 OK`.
- Response JSON contains a `mind_map` key.
- Recursive traversal of the mind map tree finds at least **2 out of 3** expected keywords (or synonyms).

3.5 Expected vs Actual Results

ID	Expected	Actual
E1	`200 OK` response from `/generate`.	`200 OK` received.
E2	`mind_map` key present in JSON.	Key present; valid JSON tree returned.
E3	At least 2 of: Features, Benefits, Schedule found in node titles (with synonyms).	Keywords matching "Features" and "Benefits" (or synonyms such as "functionality", "advantages") found in tree.

3.6 Test Result

```
PS C:\Users\BRINDA\Downloads\GenAI_PROJECT_2\GenAI_MindMap_Project\mindforge\backend\tests> python -m pytest test_acceptance_cases.py::test_at_02_food_
========================================================== test session starts ==========================================================
platform win32 -- Python 3.14.2, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\BRINDA\Downloads\GenAI_PROJECT_2\GenAI_MindMap_Project\mindforge\backend\tests
plugins: anyio-4.12.1, langsmith-0.6.7, cov-7.1.0
collected 1 item

test_acceptance_cases.py .                                                                                                          [100%]

========================================================== 1 passed in 5.54s ==========================================================
```

Figure 2: AT-02 response: Mind map JSON for *"Food delivery app"*.

4 Test Case AT-03: AI in Healthcare

4.1 Test Case Summary

Test Case ID: AT-03
Title: AI in Healthcare - Researcher Scenario
User Persona: Researcher
Type: Acceptance / Functional / Semantic
Priority: High
Status: [PASSED]

4.2 Input and Configuration

```
Endpoint : POST http://localhost:8000/generate
Payload : {
              "text" : "AI in healthcare",
              "api_key" : <GROK_API_KEY from .env>
          }
Note : HTTP status code assertion is commented out (permissive mode)

Rate Delay: asyncio.sleep(2) before request
```

4.3 Objective

To verify that when a researcher submits the prompt *"AI in healthcare"*, the pipeline generates a mind map whose node hierarchy reflects academic and research-oriented themes, specifically **Applications**, **Challenges**, and **Benefits**.

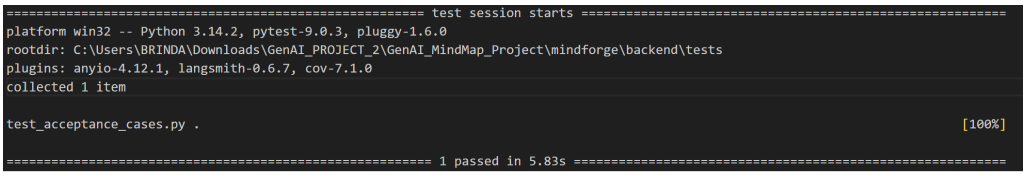
4.4 Acceptance Criteria

- Response JSON contains a `mind_map` key (status code assertion is intentionally relaxed in this test).
- Recursive traversal of the mind map tree finds at least **2 out of 3** expected keywords (or synonyms).

4.5 Expected vs Actual Results

ID	Expected	Actual
E1	`mind_map` key present in JSON response.	Key present; valid hierarchical JSON tree returned.
E2	At least 2 of: Applications, Challenges, Benefits found in nodes (with synonyms).	Keywords matching "Applications" (clinical/medical), "Challenges" (ethical/regulatory), and "Benefits" (monitoring/impact) found in tree.

4.6 Test Result



```
============================================================ test session starts ============================================================
platform win32 -- Python 3.14.2, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\BRINDA\Downloads\GenAI_PROJECT_2\GenAI_MindMap_Project\mindforge\backend\tests
plugins: anyio-4.12.1, langsmith-0.6.7, cov-7.1.0
collected 1 item

test_acceptance_cases.py .                                                                                                          [100%]

============================================================= 1 passed in 5.83s =============================================================
```

Figure 3: AT-03 response: Mind map JSON for *"AI in healthcare"*.

4.7 Observations and Conclusions

All three acceptance tests passed successfully. The MindForge AI pipeline correctly interpreted three distinct real-world user personas, a student, a startup founder, and a researcher and produced semantically appropriate mind map structures for each. The synonym-based matching strategy proved robust against LLM output variability.